Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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June 22, 2026

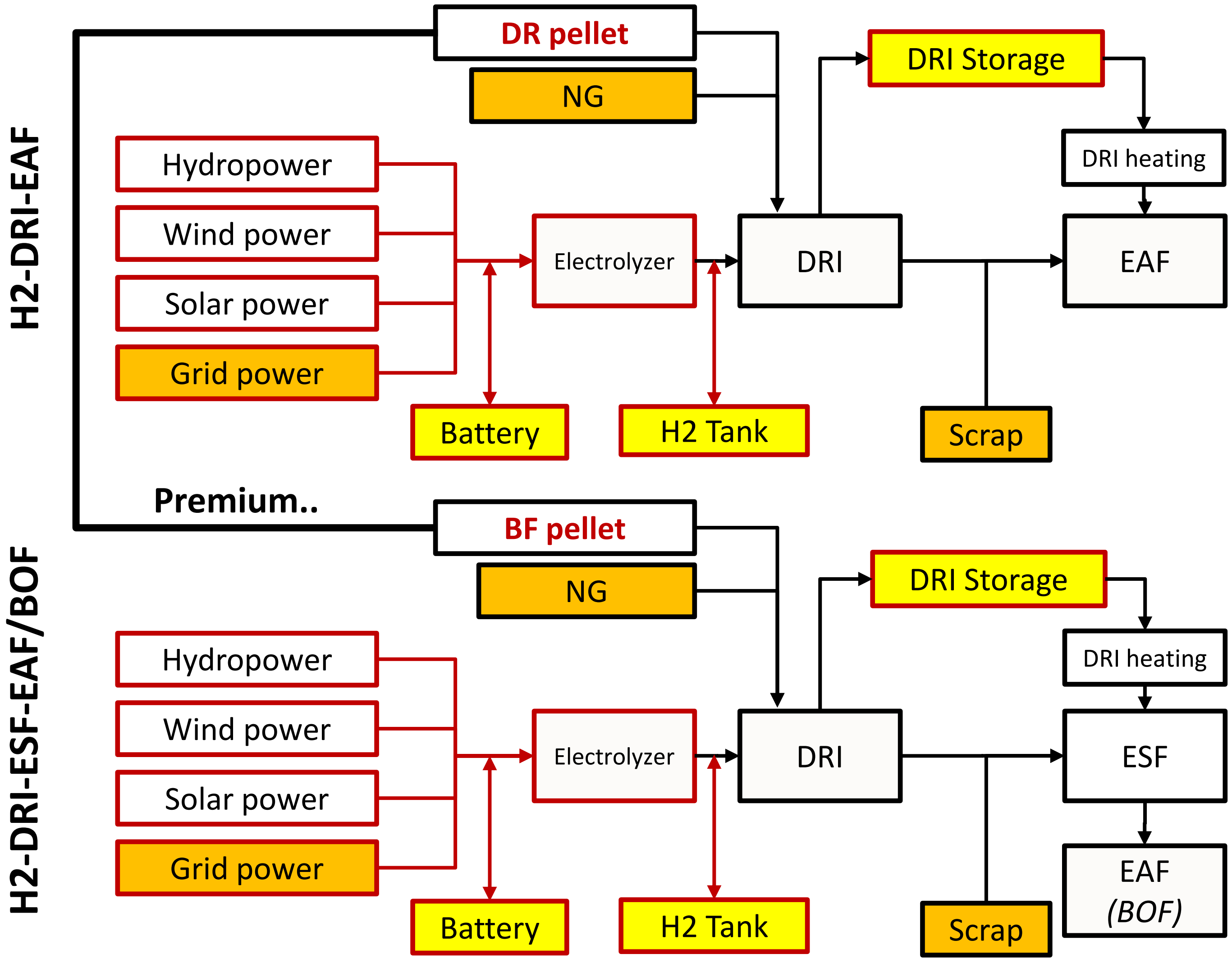
Project Timeline

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model upgrade (Base on case India)																								
Case study: India (With upgrading models)																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: World green iron trading routes																								
Final report																								

Preview of today’s presentation

- Research progress on global high-resolution green steel levelized cost distribution

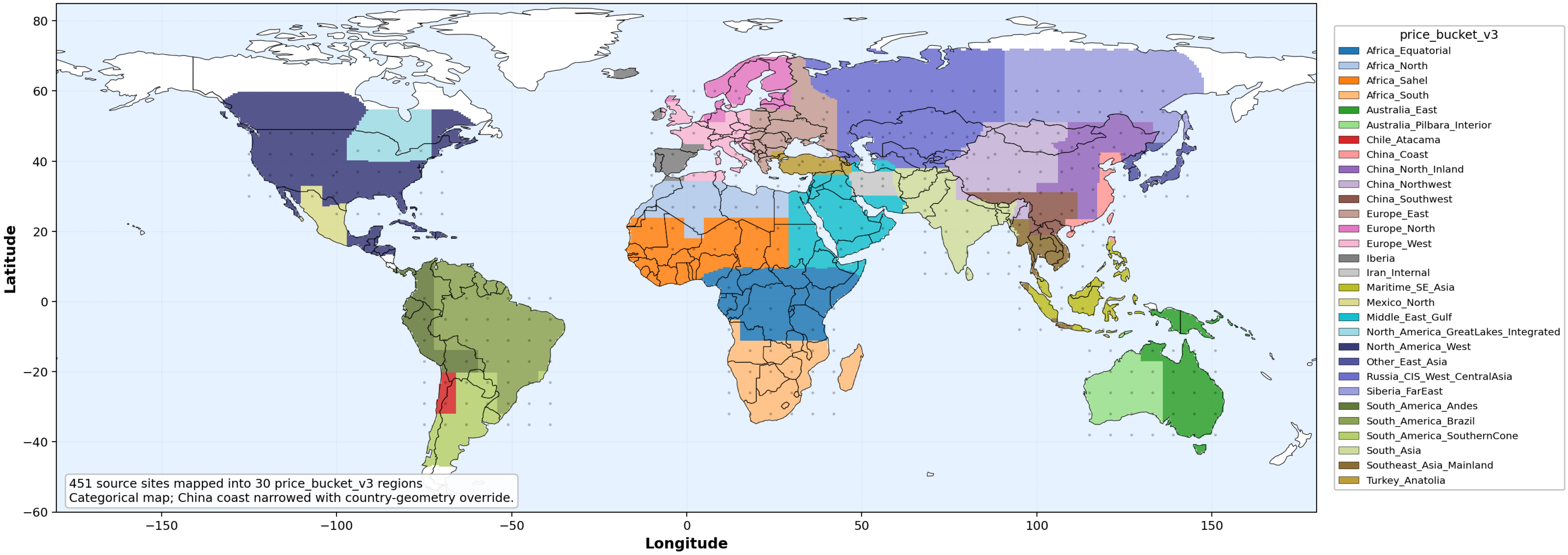
The definition: Green iron & steel technologies



- Great flexibility in every furnaces
- Expensive DR-grade iron ore demanded

- Cheaper iron ore available
- Potential inheritance of existing BOF
- *But..* additional ESF equipment CAPEX/OPEX
- Inflexible smelting furnace

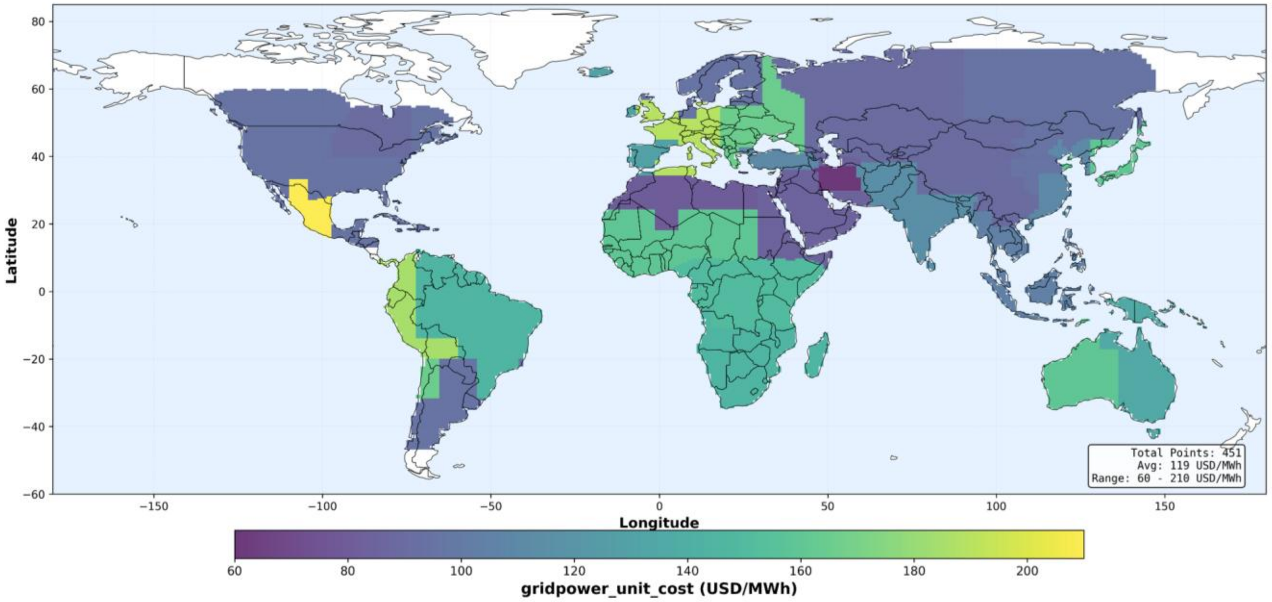
The definition: Regional division



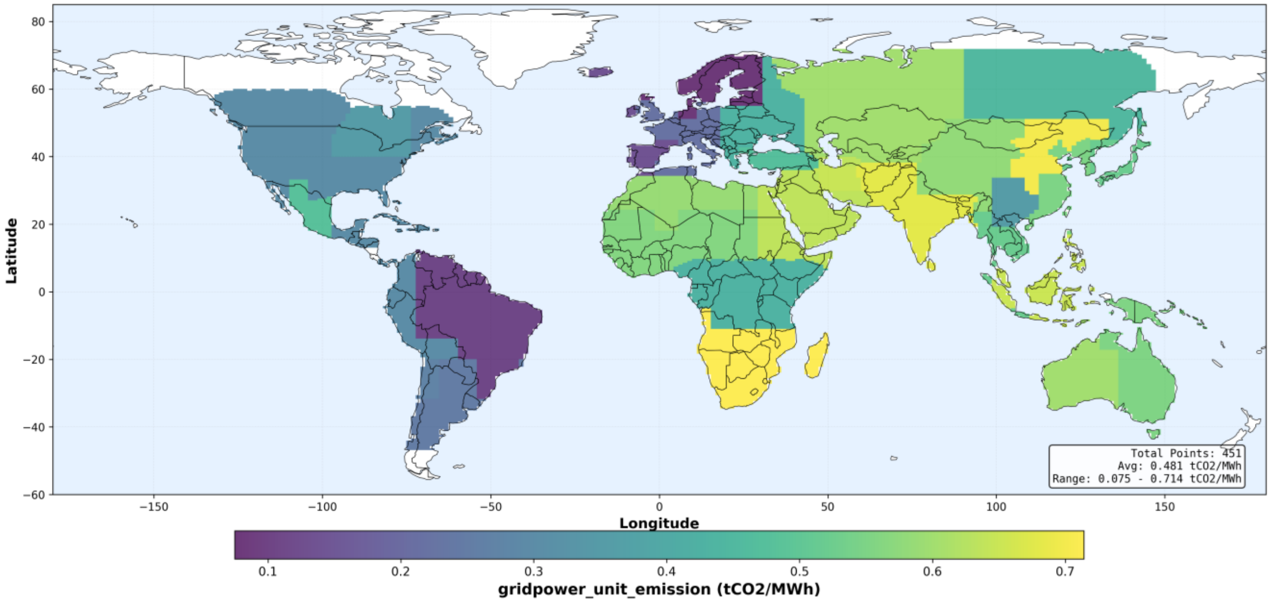
- 30-regions version, Special division of some major steel-producing areas
- Using AI mainly to investigate datasets; Primarily based on publicly available prices and market data from 2024-2026, with some technical and economic parameters and equipment capital cost assumptions derived from reports/literature from 2022-2023.

Research progress: Focus on the worldwide distribution

The definition: Grid power



Grid power cost(USD/MWh)

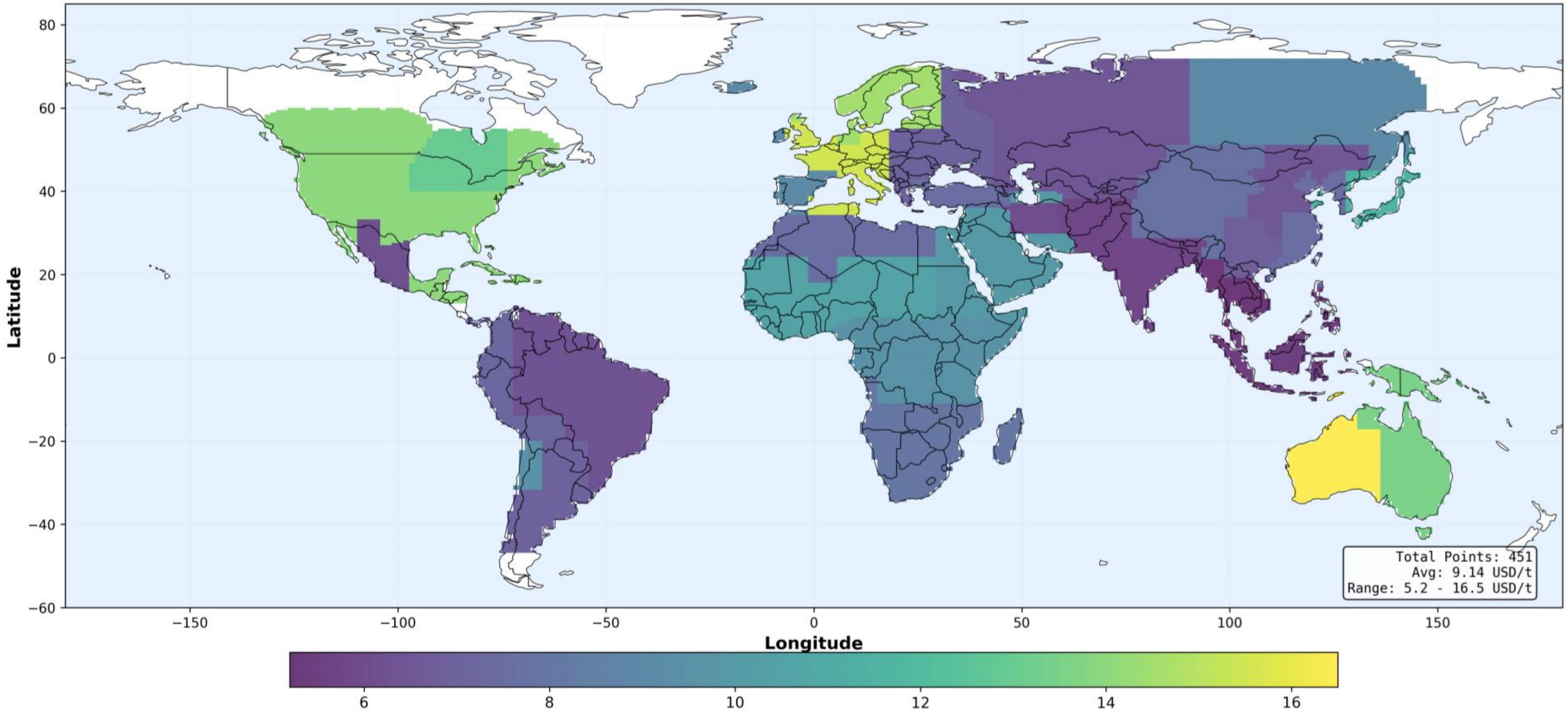


Grid emission(tCO2/MWh)

- High cost regions: Europe, Mexico(import NG-dominated), South America-Andes, Africa(inefficient infrastructure)
- High emission regions: Northern China, India, Southern Africa
- Brazil and Northern Europe have the lowest grid emission factors due to their extremely high renewable energy coverage.
- (This price mostly uses historical data from 22-23 years ago; it may need to be changed to forecast data from 2030.)

Research progress: Focus on the worldwide distribution

The definition: Labor cost



Iron unit labor cost(USD/MWh)

- Basically corresponds to the average social wage
- Africa: Due to a shortage of skilled workers needed in the steel industry, overall labor costs remain high

Research progress: Focus on the worldwide distribution

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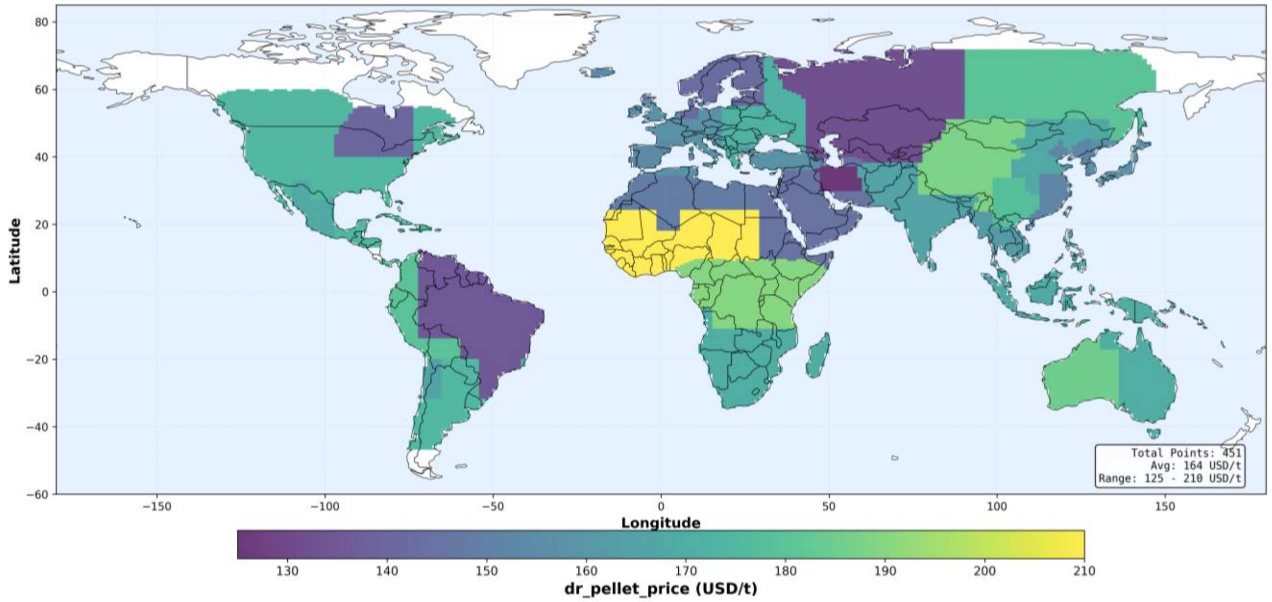
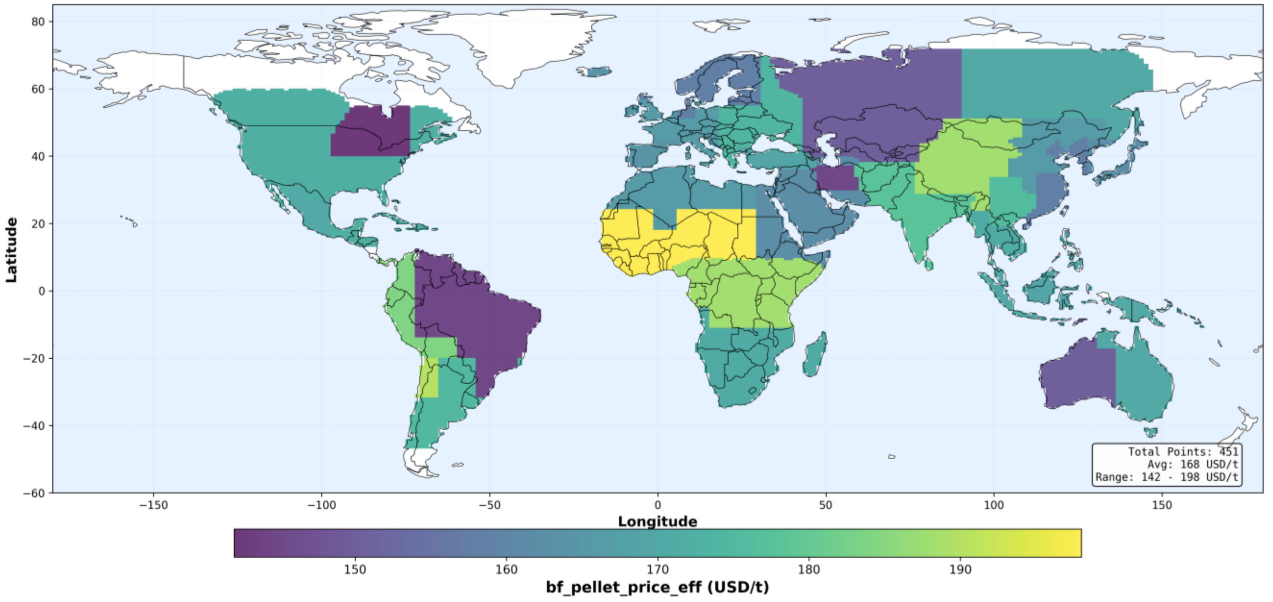
The definition: Iron ore



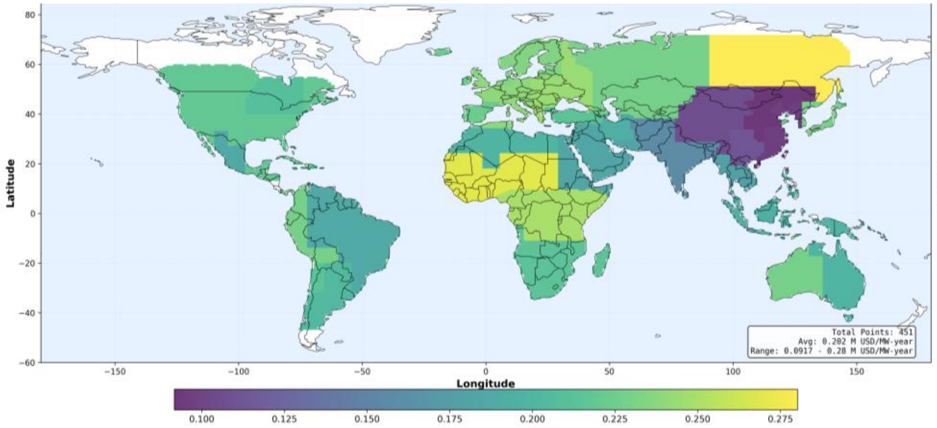
DR pellet price USD/t



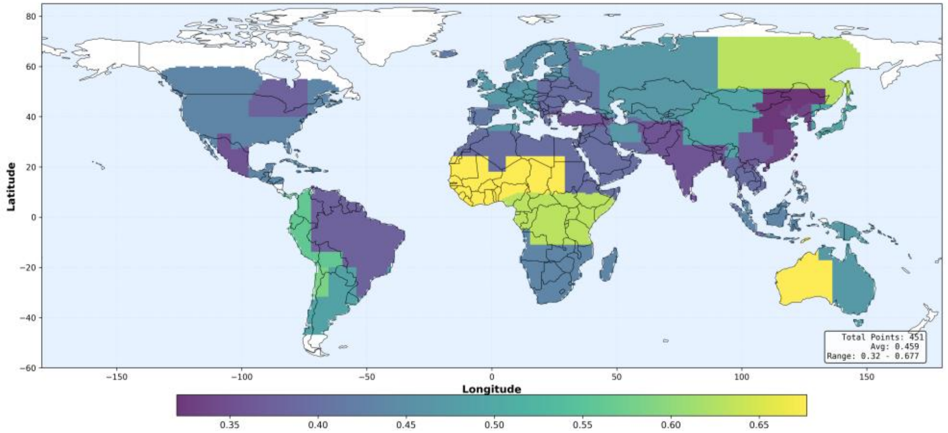
BF pellet price USD/t

- Low-price areas: Generally, resource-producing regions.
- High-price areas: Generally, areas with high demand; small-scale markets (West Africa, etc.) may experience premiums due to a lack of bargaining power.
- Australia: produce mainly BF-grade pellet, so DR-grade pellet cost relatively more.

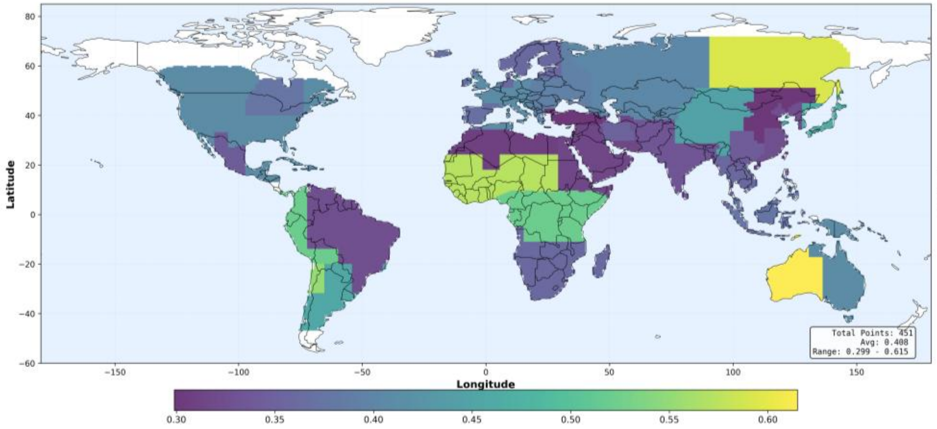
The definition: Annual costs (CAPEX/OPEX included, levelized)



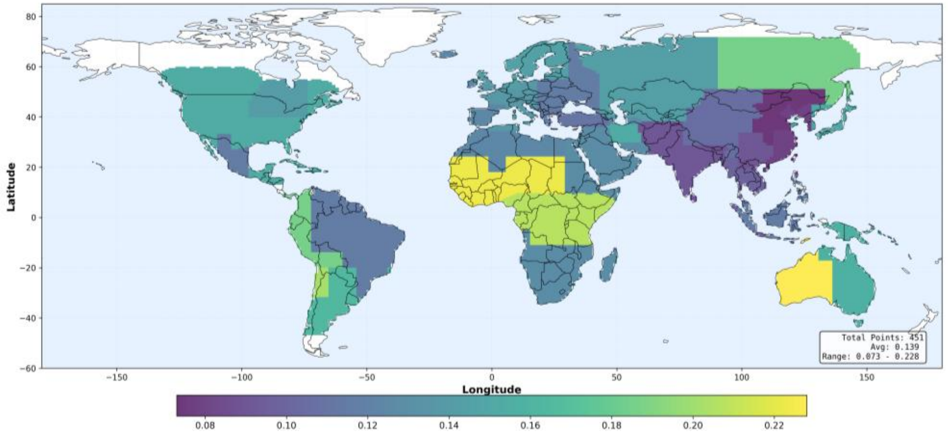
Electrolyzer MUSD/(MW/Y)



DRI furnace MUSD/(t/h)



EAF MUSD/(t/h)



BOF MUSD/(t/h)

- **Price distribution of different equipment is not uniform**: Regions with corresponding industrial and infrastructure foundations generally have lower prices.

Research progress: Focus on the worldwide distribution

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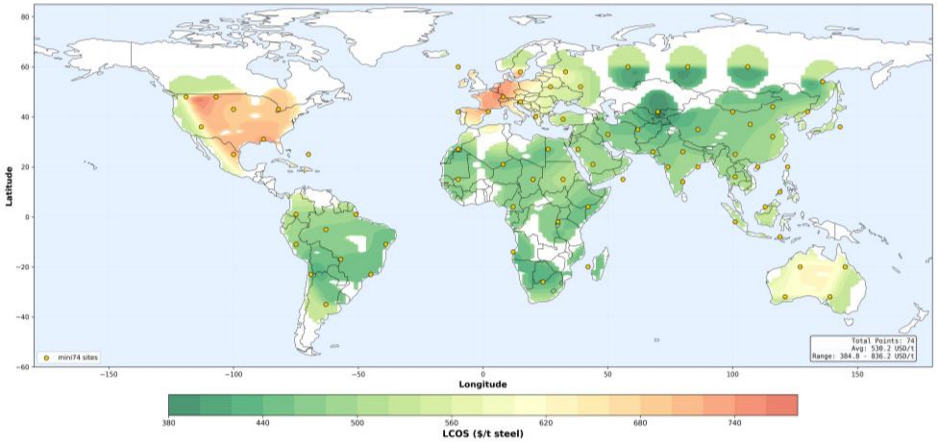
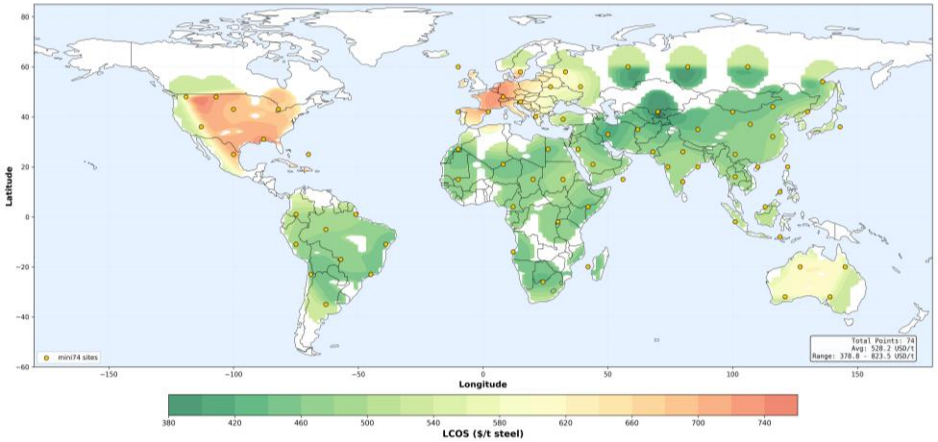
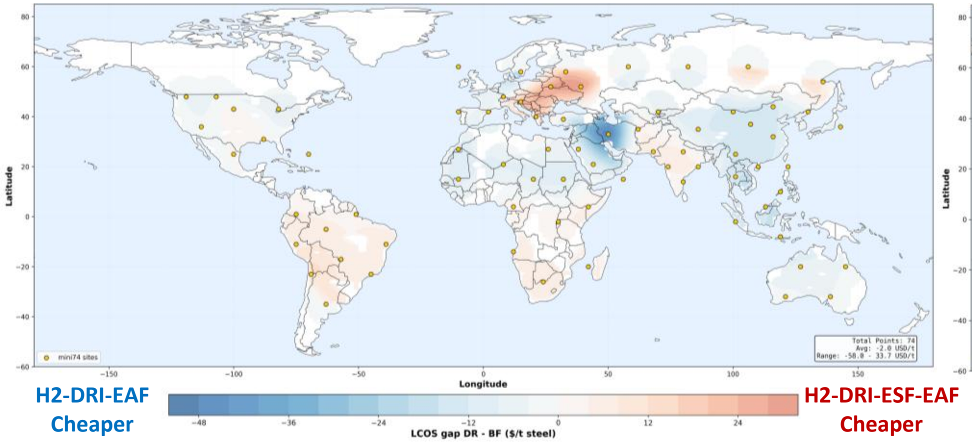
Results – (also a demo – Lightweight mode)



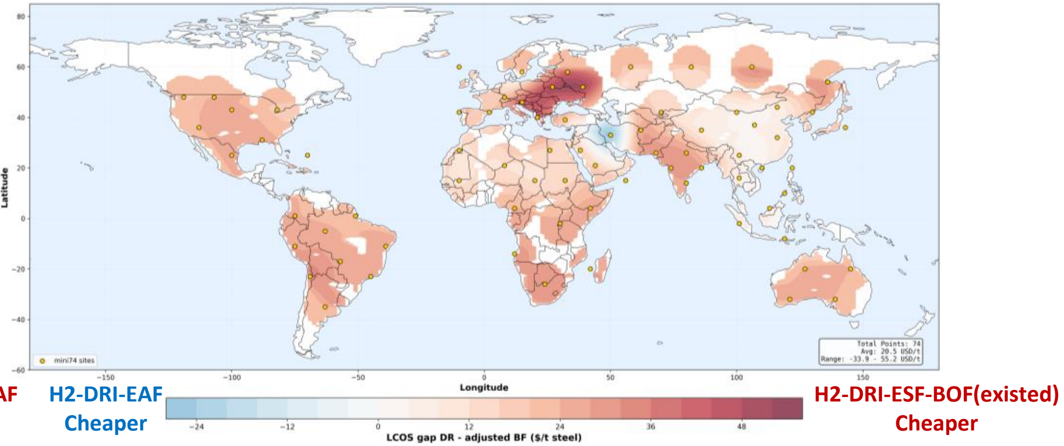
H2-DRI-ESF-EAF | LCOS (USD/tcs)



H2-DRI-EAF | LCOS (USD/tcs)



[H2-DRI-EAF]-[H2-DRI-ESF-EAF] | ΔLCOS (USD/tcs)



[H2-DRI-EAF]-[H2-DRI-ESF-BOF(existed)] | ΔLCOS (USD/tcs)

- A full-scale process (6° step size) takes about a week for a single run, so this **lightweight mode** is used for debugging (a single run only takes an hour).
- Because of the high CAPEX and high labor cost, **Europe/US/Australia shows a higher LCOS**.
- H2-DRI-EAF and H2-DRI-ESF-EAF shows different preferences in different regions, but if relying on existing BOFs for retrofitting, H2-DRI-ESF-BOF performs better in most regions.

Next steps

- Based on the discussion and research, further improve the accuracy of the data or the resolution of the region division.
- Produce full-scale high-resolution images.

Thank You For Your Attention!